

Name: _____ ID: _____ Section: _____

Question 1 [8 Points]

Suppose, you work for a gift shop called “One Price Store” and your job is to calculate the total and VAT amount for a purchase. Here, each item is worth 150 tk.

Draw a flowchart that takes the quantity of gift items as input and calculates the VAT that will be applied on the total price and then the total amount. Here, **7% VAT** is applied on the total price.

Sample Case 1	Sample Case 1
Sample Input: Gift quantity: 7	Sample Input: Gift quantity: 13
Sample Output: Vat: 73.5 tk Total amount: 1123.5 tk	Sample Output: Vat: 136.5 tk Total amount: 2086.5 tk
Explanation: Here, the price is = $7 * 150 = 1050$ VAT = 7% of 1050 = 73.5 Total price: 1123.5	Explanation: Here, the price is = $13 * 150 = 1950$ VAT = 7% of 1950 = 136.5 Total price: 2086.5

Question 2 [5 Points]

Write the output for the following operations:

Sl.	Operations	Output
1.	<code>System.out.println(35 % 12)</code>	
2.	<code>System.out.println(36 % 5 24 / 4)</code>	
3.	<code>System.out.println("Quiz (0+1)")</code>	
4.	<code>System.out.println(true && !(true false))</code>	
5.	<code>System.out.println(5 == 5 33 <= 30)</code>	

Question 3 [7 Points]

Write a Java program that takes a 3 digit number as an integer input from the user and calculates the sum of all the digits.

Sample Case 1	Sample Case 1
Sample Input: Enter a 3 digit number: 123	Sample Input: Enter a 3 digit number: 584
Sample Output: Sum: 6	Sample Output: Sum: 17
Explanation: Here, the input is = 123 Sum = 1 + 2 + 3 = 6	Explanation: Here, the input is = 584 Sum = 5 + 8 + 4 = 17